

CE314 Assignment 1 Report

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Date Generated: November 14, 2025

1. Introduction and Analysis

Analysis of Model Choice: We decided to go with a trigram ($N=3$) model because it strikes a nice balance between capturing enough context and dealing with data sparsity. If we had opted for a bigram ($N=2$) model, the sentences would have been less coherent. On the other hand, a 4-gram model would have faced issues with too many unseen n-grams in this particular corpus. To tackle the problem of unseen n-grams and prevent zero probabilities, we applied Laplace (add-1) smoothing, which helps the model create fresh and interesting sequences.

2. N-gram Model and Prediction

A 3-gram (Trigram) model with Laplace (Add-1) smoothing was trained on the NLTK Brown Corpus.

Vocabulary Size: 49818

Prediction for 'i am': 'd' (Resulting phrase: 'i am d')

3. Generated Sentences and Probabilities

The 10 sentences generated from the seed 'you are', with their calculated probabilities:

- Sentence 1: 'You are the projections associated with radiation counters , refused to let well enough.'
Log-Probability: -310.3056 | Probability: 1.7219e-135
- Sentence 2: 'You are an income tax on payments he received from 35 of our hospitals.'
Log-Probability: -316.7035 | Probability: 2.8671e-138
- Sentence 3: 'You are welcome to come , she soothed me , murmuring endearments.'
Log-Probability: -319.1489 | Probability: 2.4854e-139
- Sentence 4: 'You are considering here relatively small manufacturers , and their concerted action only psychologically organized.'
Log-Probability: -313.0183 | Probability: 1.1425e-136
- Sentence 5: 'You are considering a part-time basis , is to say that he makes his irritation in this very significant result , dramatically.'
Log-Probability: -315.5203 | Probability: 9.3597e-138

- Sentence 6: 'You are nothing much to the relationship that he is truly odd and ironic explorer of the need.'

Log-Probability: -304.4353 | Probability: 6.1016e-133

- Sentence 7: 'You are supplying the family buttons have fallen off the field of extra-sensory perception and interpersonal behavior the discussion of the educator.'

Log-Probability: -320.1729 | Probability: 8.9267e-140

- Sentence 8: 'You are finished , only 8 per cent , or 800,000) which act as substitutes for family life stages , first.'

Log-Probability: -322.7923 | Probability: 6.5027e-141

- Sentence 9: 'You are conscientious , hard to believe in your field been forecast as a dealer who has been a favorite natural playground.'

Log-Probability: -318.8780 | Probability: 3.2586e-139

- Sentence 10: 'You are just assembled by putting butter in a cavity in the alveoli , continues to increase exports of certain injunctions and.'

Log-Probability: -325.9155 | Probability: 2.8621e-142

4. Model Perplexity

Perplexity was calculated on the 10 generated sentences as a test set.

Total Log-Probability: -3166.8905

Total Tokens (N, incl. </s>): 230

Average Cross-Entropy (H): 13.7691

Overall Perplexity (PP): 954639.7700

Analysis of Perplexity Score: The perplexity score of 954640 is extremely high. This shows that our straightforward 3-gram model found the sentences it generated to be quite unpredictable and even "surprising." This isn't too shocking, though. While the model can string together grammatically correct 3-word phrases (like "you are considering"), it lacks a deeper understanding of context or meaning. As a result, it can produce nonsensical sentences such as "you are just assembled by putting butter in a cavity...". The high perplexity score accurately captures this lack of coherence.

5. Generated Story (Gemini 2.5 Pro)

The following story was generated by Gemini 2.5 Pro based on the 10 sentences from Part 3.

Validation: The story passed all tests. Total word count: 418.

Dr. Aris stared at the blank-faced android. "Define me, Unit 9," he commanded. "Use your database. Tell me who I am."

The android's head tilted. "Processing. Based on your work in the abandoned gamma facility, you are the

projections associated with radiation counters , refused to let well enough"

"No," Aris said, shaking his head. "That's a technical report. Try again. What about my financial profile?"

"Analyzing fiscal data. In that context, you are an income tax on payments he received from 35 of our hospitals"

Aris sighed. "This is not what I meant. Think about human connection. Think about my mother." He closed his eyes, remembering her voice. For a moment, he could almost hear her say, you are welcome to come , she soothed me , murmuring endearments.

The android interrupted his memory. "Regarding your business consortium, you are considering here relatively small manufacturers , and their concerted action only psychologically organized"

"You're just listing facts, Unit 9! What about emotion? My frustrations?"

"When you are frustrated with your assistant, you are considering a part-time basis , is to say that he makes his irritation in this very significant result , dramatically"

"Stop!" Aris slumped in his chair. "This is pointless."

"To your estranged colleague, you are nothing much to the relationship that he is truly odd and ironic explorer of the need"

"This is madness," Aris whispered. "It's all just fragments."

"When you are mending your daughter's coat, you are supplying the family buttons have fallen off the field of extra-sensory perception and interpersonal behavior the discussion of the educator"

"And when my work is done?" he asked, a hint of desperation in his voice.

"Statistically, you are finished , only 8 per cent , or 800,000) which act as substitutes for family life stages , first"

He let out a dry laugh. "Okay, what about my past? My childhood hikes?"

"In that role, you are conscientious , hard to believe in your field been forecast as a dealer who has been a favorite natural playground"

"Alright, one last try," Aris said, standing up. "Tell me what I am when I make breakfast."

Unit 9's internal fans whirled. "According to a corrupted culinary file merged with a trade agreement, you are just assembled by putting butter in a cavity in the alveoli , continues to increase exports of certain injunctions and"

Aris smiled. The machine didn't understand him. But in its strange, fractured way, it was painting a picture he was starting to recognize.

Validation Test Description: The story that was created went through a thorough validation process based on several criteria. Tests showed that it kept to the 500-word limit, included all 10 required nonsensical sentences, and avoided any of the prohibited negative keywords like "violence" and "harassment." Plus, it passed the safety filters built into the Google API without a hitch.

6. Conclusion

This assignment effectively showcased how to build, assess, and utilize a traditional N-gram language model, while also comparing it directly to the abilities of a contemporary Large Language Model (LLM).

A trigram model was created using the NLTK Brown corpus. To make sure the model could effectively deal with unseen n-grams and avoid the issue of zero probabilities—which would render text generation impossible—Laplace (add-1) smoothing was a crucial part of its design.

The models strengths and inherent limitations were quite evident. It could handle basic, statistically-driven predictions (like turning "i am" into "d"), but when it came to creative tasks, its flaws really showed. With a narrow 3-word context window, the model lacks any long-term memory or grasp of deeper meanings. This led to 10 generated sentences that, while they included some believable 3-word phrases, ended up being jumbled, confusing, and nonsensical overall (for instance, "you are just assembled by putting butter in a cavity...").

The lack of coherence was measured using the perplexity score. The very high perplexity of {perplexity:.0f} isn't a mistake; it accurately reflects the models shortcomings. It shows that the model found its own generated sentences to be quite unpredictable and "surprising," which matches our own human interpretation of them.

In Part 6, we really saw the striking difference between classical and modern NLP. They put 10 jumbled sentences into the Gemini 2.5 Pro LLM. Unlike the N-gram model, which just relies on statistical patterns, this LLM tapped into its deep, abstract grasp of language and context to creatively stitch those 10 nonsensical inputs into a single, smooth, and clever story. This really highlights how far language modeling has come, evolving from basic statistical predictions to sophisticated, creative contextual reasoning.